

ENGR 102 Summer Bridge

Day 16 Student Quiz

Indexed Mutation and Independent Exam 2 Rehearsal

20 points • approximately 20-25 minutes

Name		UIN		Section	
------	--	-----	--	---------	--

Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace indexed mutation and post-mutation state.

4 points

```
data = [2, 11, -3, 8]
edits = 0
for index in range(len(data)):
    if data[index] < 0:
        data[index] = 0
        edits = edits + 1
    elif data[index] > 10:
        data[index] = 10
        edits = edits + 1
print(data, edits)
```

Give list state after every pass and exact output.

Question 2. Repair non-mutating code.

2 points

```
data = [-2, 5]
for value in data:
    if value < 0:
        value = 0
print(data)
```

Explain why the list is unchanged and give the essential loop repair.

--

Question 3. Program construction

14 points

Write one complete Python program that satisfies every requirement.

- Use measurements = [9, -1, 12, 21, 6].
- Mutate negative values to 0 and values above 15 to 15.
- Count modified positions and compute the total from final values.
- Print final list, modified count, and total on separate lines.
- Do not use sum or a comprehension. Give one test in which both the first and last positions require changes.

[illegible]

Test evidence / expected result